Caesar & Vigenère Cipher

Implementation · Cryptanalysis · Security Analysis

CLASSICAL ENCRYPTION

CCE 4236 / CCE 442

Tanta University — Faculty of Engineering

Submitted to **Dr. Tahani Allam**

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What Is This Project?

Two classical ciphers implemented from scratch in Python with a full interactive Streamlit dashboard. The goal: not just to build them — but to **break them**, and understand why classical encryption is fundamentally insecure by modern standards.

Caesar Cipher

Fixed-shift monoalphabetic substitution

Vigenère Cipher

Keyword-based polyalphabetic substitution

Brute Force Attack

All 25 keys enumerated and ranked

Frequency Analysis

Chi-squared statistical cryptanalysis

File Encryption Mode

Upload, process, and download .txt files

Caesar Cipher — The Fixed Shift

Every letter shifts forward by a fixed number of positions in the alphabet. The same key encrypts and decrypts. Non-alphabetic characters pass through unchanged; case is preserved.

Formulas

Encrypt:

$$C = (P + \text{key}) \bmod 26$$

Decrypt:

$$P = (C - \text{key} + 26) \bmod 26$$

Worked Example: `HELLO` → Key 3 → `KHOOR`

Plaintext	H	E	L	L	O
Position	7	4	11	11	14
+ Key (3)	10	7	14	14	17
Ciphertext	K	H	O	O	R

Vigenère Cipher — The Keyword Shift

A keyword defines a **different shift for each letter**, repeating to cover the full message. This polyalphabetic approach was considered unbreakable for nearly 300 years.

Formulas

Encrypt:

$$C_i = (P_i + K_i) \bmod 26$$

Decrypt:

$$P_i = (C_i - K_i + 26) \bmod 26$$

Worked Example: `ATTACK` + Keyword `KEY` → `KXRKGI`

Plaintext	A	T	T	A	C	K
Keyword	K	E	Y	K	E	Y
Ciphertext	K	X	R	K	G	I

Key Insight: The two T's encrypt to **X** and **R** — same letter, different result. This destroys the statistical fingerprint that makes Caesar trivially breakable.

Caesar Is Cryptographically Dead — And Vigenère Is Not Far Behind

Why Caesar Fails

1

Tiny Keyspace

Only 25 possible keys. Brute force takes milliseconds.

2

Frequency Fingerprint Survives

Every 'E' always maps to the same ciphertext letter. One frequency count recovers the key.

3

No Diffusion

Each character encrypted independently. Plaintext patterns mirror ciphertext patterns.

Security Comparison

Property	Caesar	Vigenère
Type	Monoalphabetic	Polyalphabetic
Keyspace	25 keys	26^n keys
Same letter → same cipher?	Always	No
Frequency analysis	Trivially broken	Resists naive attack
Brute-forceable?	25 attempts	Infeasible (long keyword)

25 keys vs AES-256's 2^{256} — the comparison is not even close.

Brute Force — All 25 Keys, Ranked

Since Caesar has only 25 keys, we try them all. Each candidate decryption is scored against English letter frequencies using **chi-squared distance**. The lowest score wins.

Attack Results — Ranked by Confidence

Rank	Shift	Decrypted Text	Confidence
1	3	`Hello, World!`	94.2%
2	16	`Uryyb, Jbeyq!`	12.1%
3	9	`Axeeh, Phkew!`	8.4%

Chi-squared scoring: $\chi^2 = \Sigma\,(\text{observed} - \text{expected})^2 / \text{expected}$ — measures how closely the decrypted text matches English letter distribution.

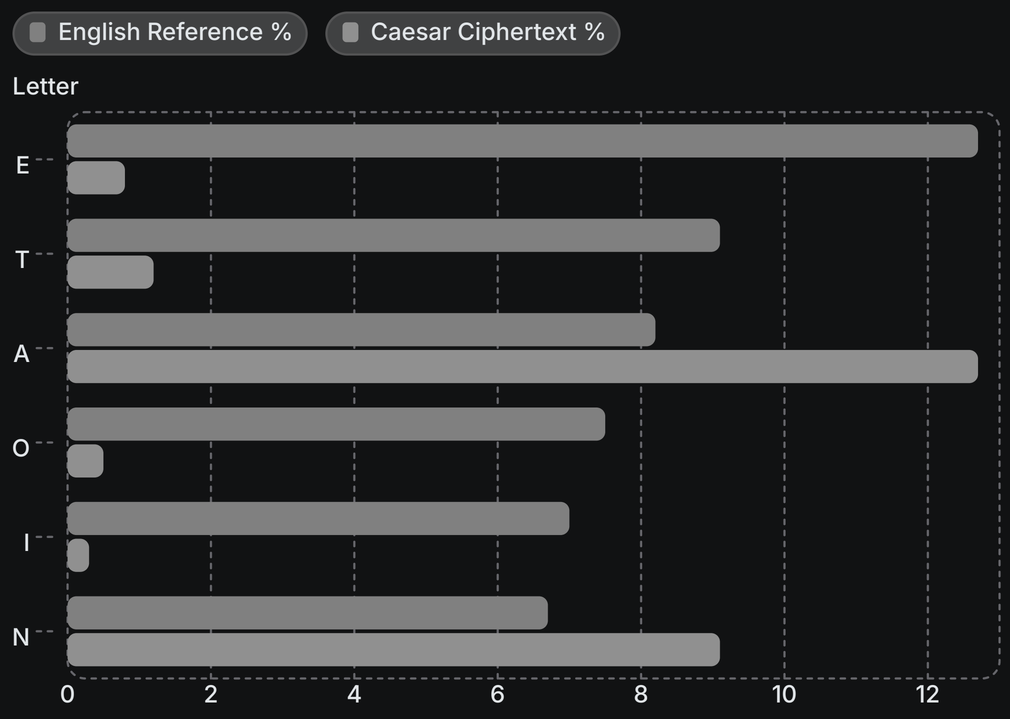
Frequency Analysis — Let the Statistics Talk

English has a known letter distribution — **E ≈ 12.7%, T ≈ 9.1%, A ≈ 8.2%**. Caesar preserves this distribution (just shifted). The offset between the ciphertext peak and 'E' reveals the key instantly.

Chi-Squared Formula

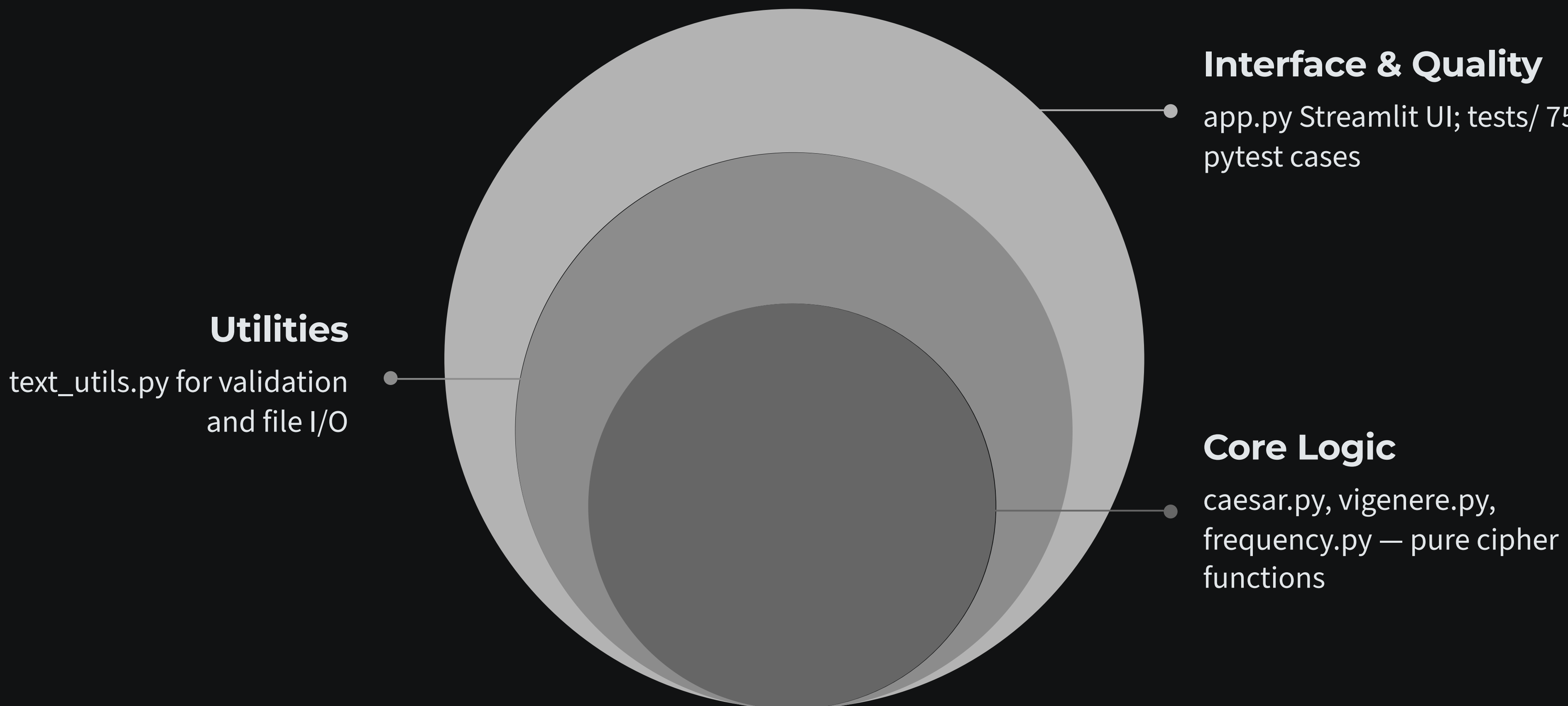
$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

Low χ^2 = close match to English = correct key.



Vigenère note: Full-ciphertext frequency analysis shows near-uniform distribution — the fingerprint is hidden. The **Kasiski test** is needed to find keyword length first, then each positional group reduces to Caesar.

How It's Built — Architecture



Core cipher logic is completely independent of the UI — pure functions, fully testable in isolation.

Python 3.9+	**Streamlit**
Plotly	**pytest**

Streamlit Dashboard — 5 Interactive Pages

Caesar Cipher

Encrypt/decrypt with live shift slider and instant output

Vigenère Cipher

Keyword input with alignment table and per-letter shift display

Brute Force

Attack panel with ranked results and chi-squared confidence scores

Frequency Analysis

Chi-squared scoring with interactive letter distribution charts

File Mode

Upload .txt files, process, and download encrypted or decrypted results

Key Takeaways

1	**Classical ciphers are historically interesting — and completely broken.** Caesar's 25-key space and Vigenère's statistical weaknesses make both trivial targets for modern cryptanalysis.
2	**Security comes from keyspace size, diffusion, and confusion.** AES-256's 2^{256} keys, combined with multi-round diffusion, is what makes modern encryption hold.
3	**Statistics are an attacker's best friend against weak ciphers.** Frequency analysis and chi-squared scoring turn language patterns into a decryption weapon.

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Built with Python · Streamlit · pytest · and a respect for the history of cryptography.